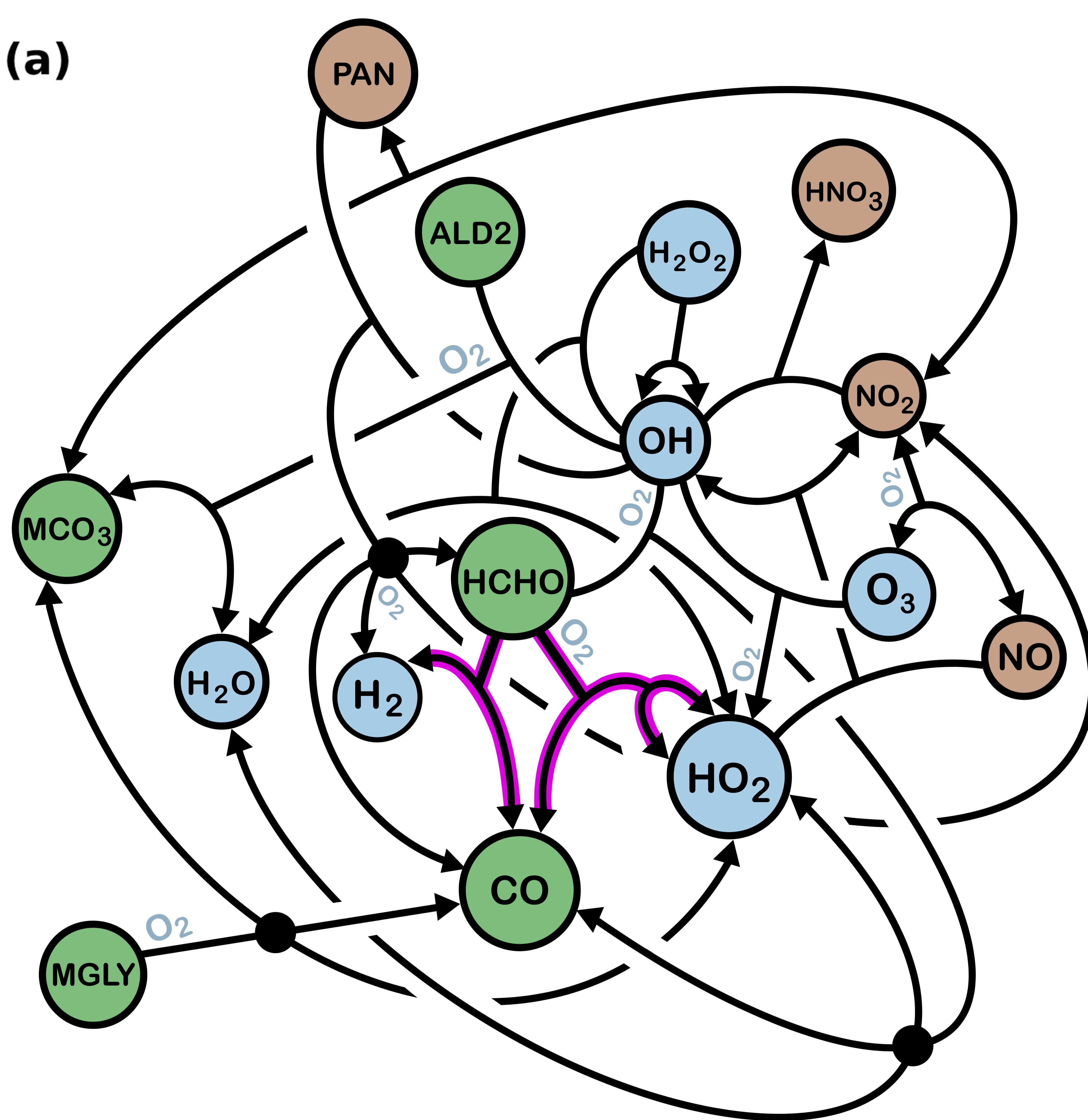


(a)

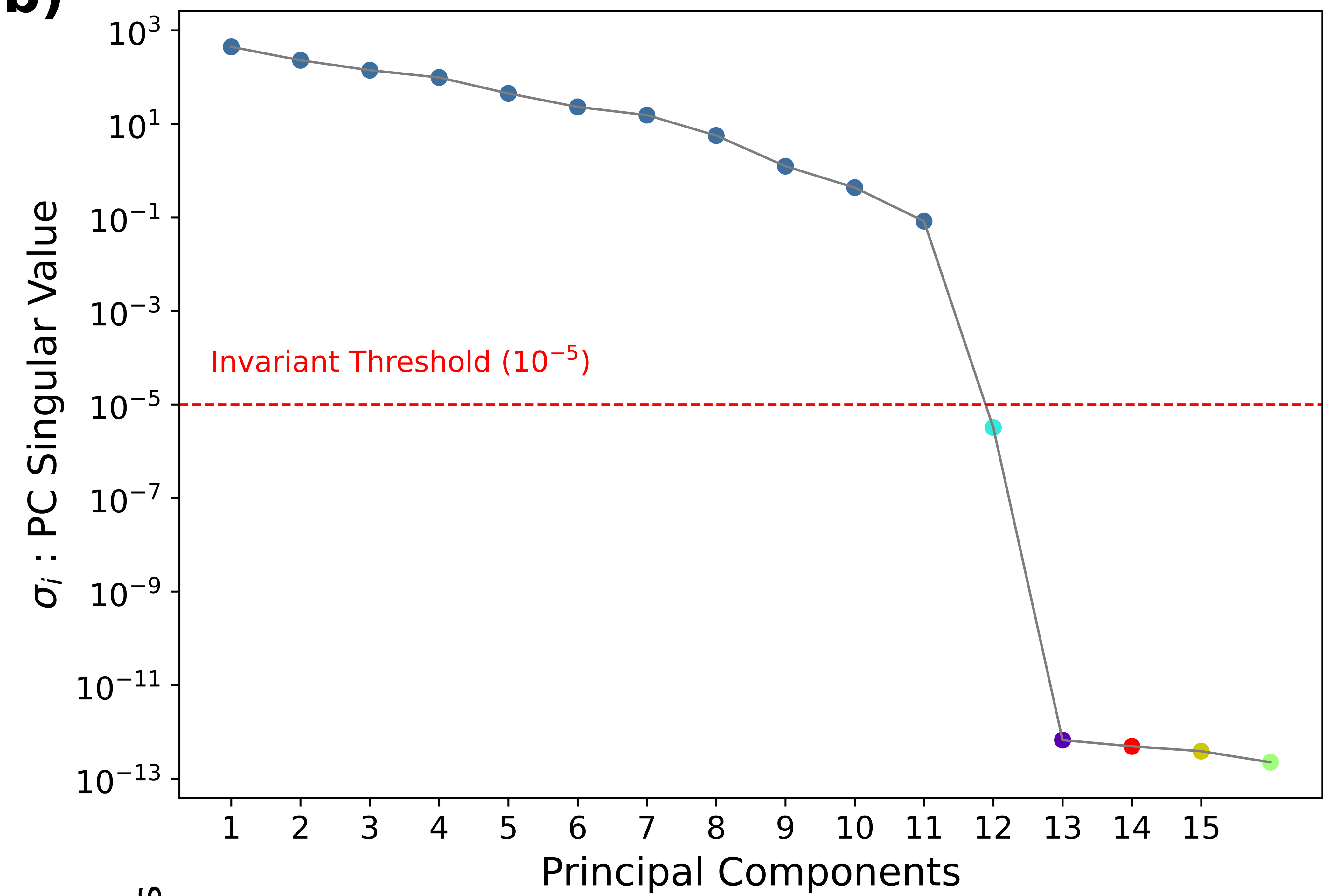
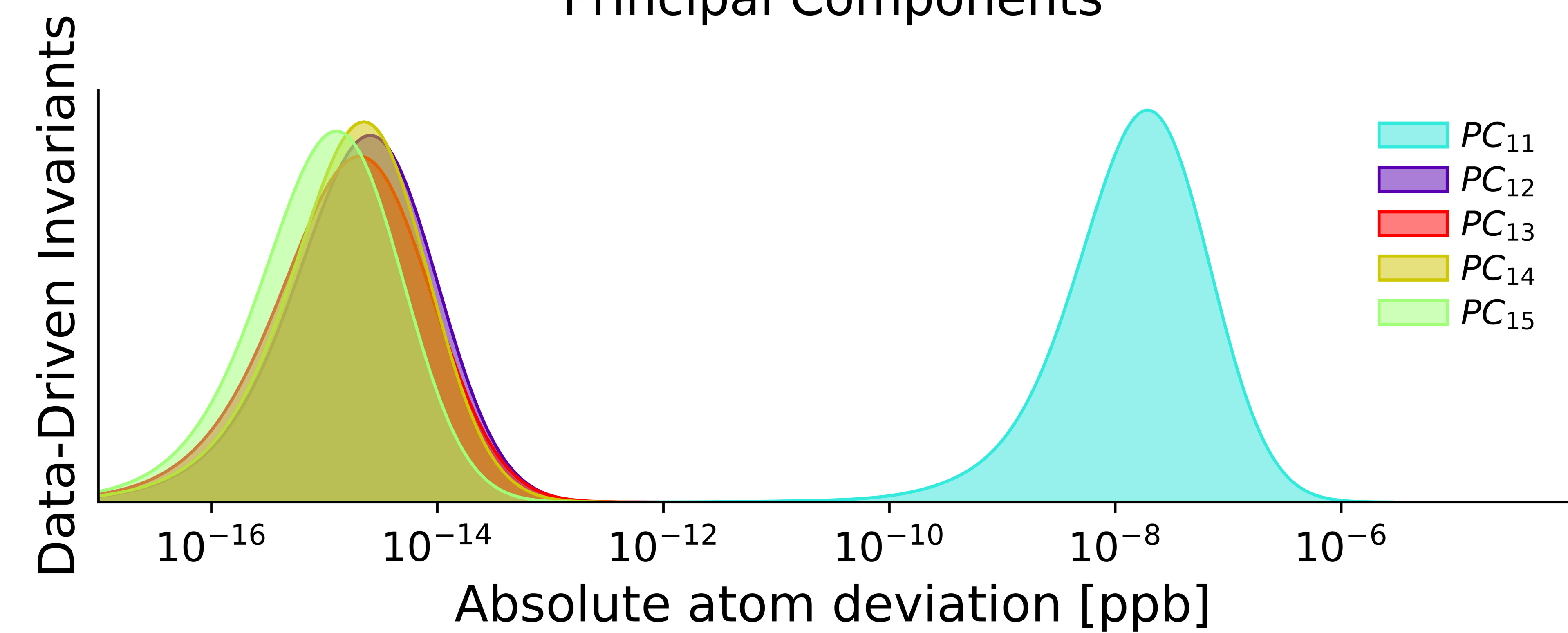
$$0 = d_t[\text{HCHO}] + d_t[\text{CO}] + 2d_t[\text{ALD2}] + 3d_t[\text{MGLY}] + 2d_t[\text{MCO}_3] + 2d_t[\text{PAN}]$$

$$0 = d_t[\text{NO}] + d_t[\text{NO}_2] + d_t[\text{HNO}_3] + d_t[\text{PAN}]$$

$$0 = 2d_t[\text{HCHO}] + d_t[\text{HO}_2] + 2d_t[\text{H}_2\text{O}_2] + d_t[\text{OH}] + d_t[\text{HNO}_3] + 2d_t[\text{H}_2] + 4d_t[\text{ALD}_2] + 4d_t[\text{MGLY}] + 3d_t[\text{MCO}_3] + 3d_t[\text{PAN}] + 2d_t[\text{H}_2\text{O}]$$

$$0 = 3d_t[\text{O}_3] + d_t[\text{NO}] + 2d_t[\text{NO}_2] + d_t[\text{HCHO}] + 2d_t[\text{HO}_2] + 2d_t[\text{H}_2\text{O}_2] + d_t[\text{OH}] + 3d_t[\text{HNO}_3] + d_t[\text{CO}] + d_t[\text{ALD2}] + 2d_t[\text{MGLY}] + 3d_t[\text{MCO}_3] + 5d_t[\text{PAN}] + d_t[\text{H}_2\text{O}] + 2d_t[\text{O}_2]$$

$$0 = -\alpha_4 d_t[\text{O}_3] + \alpha_4 d_t[\text{NO}] - \alpha_4 d_t[\text{HO}_2] + 2\alpha_3 d_t[\text{H}_2] - 2\alpha_4 d_t[\text{ALD2}] + \alpha_4 d_t[\text{PAN}] + \alpha_4 d_t[\text{H}_2\text{O}] + \alpha_4 d_t[\text{MCO}_3]$$

(b)**(c)****(d)**